MATE 664 L06
Jan-21 2026

Recap

① Diffusivity definitions
- $D_i^* \rightarrow$ self-diffusivity
- $D_i \rightarrow$ intrinsic
- $\tilde{D} \rightarrow$ interdiffusivity

②
$$D_i^* = k_B T\left(\frac{L_{11}}{C_1} - \frac{L_{1i^*}}{C_{i^*}}\right)$$

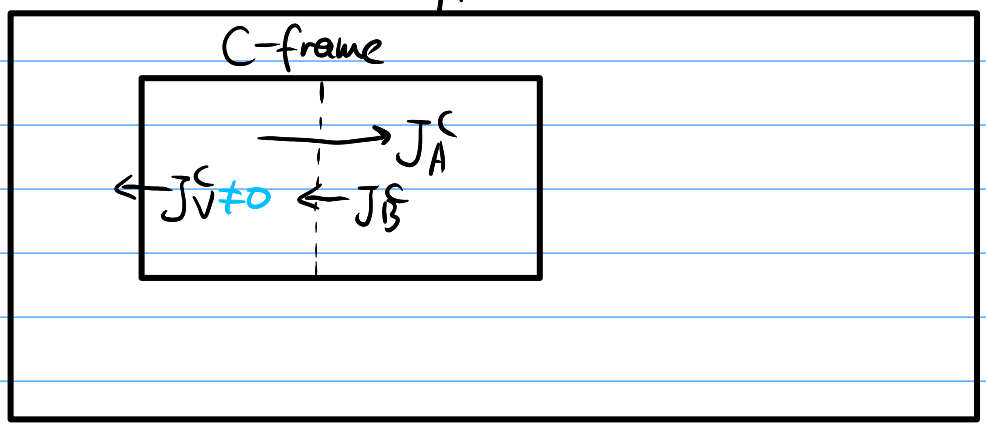
$$D_i = k_B T\left(\frac{L_{11}}{C_1} - \frac{L_{12}}{C_2}\right)\underbrace{\left(1 + \frac{\partial \ln \gamma_i}{\partial \ln C_i} + \frac{\partial \ln \langle \Omega \rangle}{\partial \ln C_i}\right)}_{\Phi(C_i)}$$

$$\tilde{D}_{12} = D_1 C_2 \Omega_2 + D_2 C_1 \Omega_1$$

Lattice C-frame / Lab V-frame both fall into Fick's equation!

Remaining questions

① Kirkendall effect how to understand?



V-frame $J_A^V = -J_B^V$

$J_V^V = 0$

Kirkendall effect

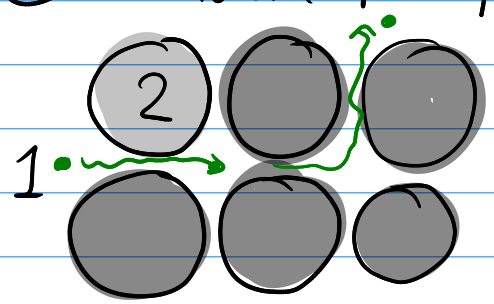
(see Mehrer & Stolwijk Diffusion Fundamentals 2009, 11, 1-32

1) Experimentally measured lattice drift

2) $J_V^c \neq 0$ But $J_V^V = 0$ ⇒ accumulation of defects in controlled volume

↳ effects matter a lot for solids !

② Interstitial diffusion



1 diffuse within sites of 2 (interstitial)

2 atoms are much heavier than 1 !

C-frame: $v_1^c = -M_1 \cdot \nabla \mu_1$ ← mobility

$$= \frac{-M_1 k_B T}{C_1} \nabla C_1 \quad \text{(dilute)}$$

$$\boxed{\begin{aligned} D_1 &= M_1 k_B T \\ &= L_{11} \frac{k_B T}{C_1} \end{aligned}}$$ ⇐ Nernst-Einstein eqn.

V-frame (similar to "diffusion through stagnant" in mass transfer)

Diffusivity of $2 \to 1 = 0$ $D_2 = 0$

$$v = (D_1 - 0)\,\Omega_1 \frac{\partial c_1}{\partial x}$$

$$J_1^v = -D_1 \frac{\partial c_1}{\partial x} + c_1 \cdot (D_1 - 0) \cdot \Omega_1 \frac{\partial c_1}{\partial x}$$

$$= -\boxed{D_1 \cdot (1 - c_1 \Omega_1)} \frac{\partial c_1}{\partial x}$$

$$\Downarrow$$

$$\tilde{D} = D_1 \cdot \Omega_2 \cdot c_2$$

$$= D_1 \cdot X_2 \quad \text{(also position-dependent)}$$

Coupling between diffusion & other D.F.s

We've discussed only $\mu_1, \mu_2 \cdots \mu_{N_i}$ as the potential
there are many other D.F.s for diffusion !

KOM ch 3.5.4 has a general comment of
"diffusion potential"

Let it be Φ_j

$$\boxed{\mu_j} + \boxed{\text{additional constraints}} + \boxed{\text{higher-order}}$$

Chemical potential

$-\mu_v \Rightarrow$ vacancy mechanism

$+q_j \phi \Rightarrow$ electromigration

$+\Omega_j P \Rightarrow$ external pressure moving sites

$+\gamma k \Omega_j \Rightarrow$ capillary

$+ \cdots\cdots$

Site volume

Surface energy

How do we calculate D?

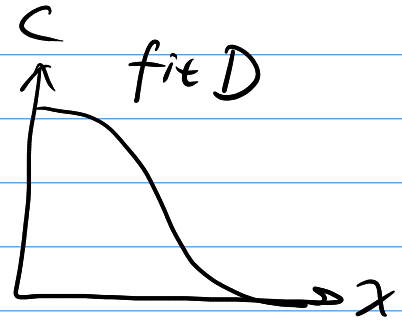
① General experiment

$\Rightarrow$ { Initial concentration
Defined geom

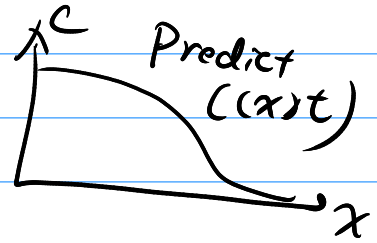
$t>0$ { measure profile / length / ....
fit back to some model $\Rightarrow D$

known Geom
known Boundary/init Cond

D unknown $\rightarrow$



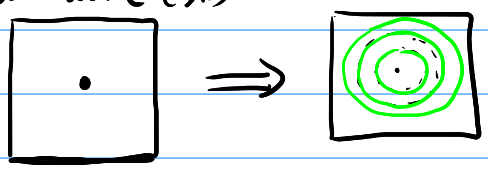
D known $\rightarrow$



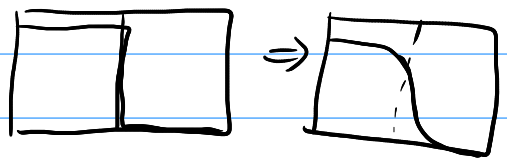
We want to understand analytical & numerical solutions to Fick's eqns.

First let's see a few experimental setups

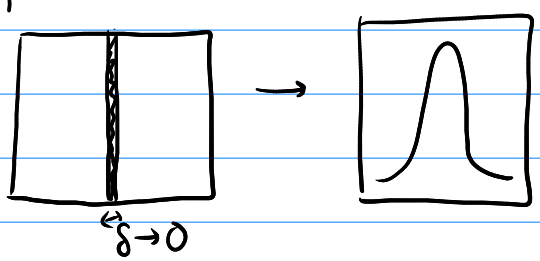
point source (3D)

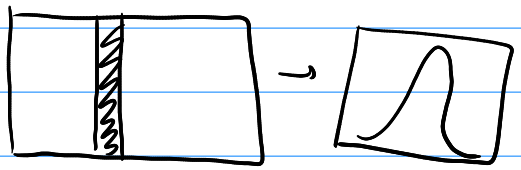


half space (semi infinite)



plane source 1D





General equation = Fick's 2nd law, assuming $\begin{cases} \text{homogeneous } D \\ \text{isotropic} \end{cases}$

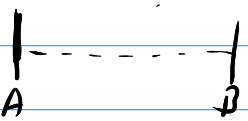
$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} \quad (1D)$$

$\hookrightarrow \begin{cases} \text{PDE} \Leftarrow c \text{ dependent on } c, x \\ \text{Linear} \Leftarrow \text{no cross term } c \cdot \frac{\partial c}{\partial x} \text{ etc} \cdots \end{cases}$

How to have unique solution?

1) Initial condition ($c(x, t=t_0) \Rightarrow$ array)

2) Boundary conditions



1° constant c at boundary ?

2° constant flux ?

We will start from steady state first

Part I Steady state analysis

$$\frac{\partial c}{\partial t} = D \nabla^2 c = 0 \Rightarrow \nabla^2 c = 0 \quad \text{Laplace equ}$$

Laplace operator depends on coordinate !

1D case

$$\nabla^2 c = \frac{d^2 c}{dx^2} = 0$$

$$\frac{d\left(\frac{dc}{dx}\right)}{dx} = 0$$

$$\int d\left(\frac{dc}{dx}\right) = \int 0 \, dx$$

$$\frac{dc}{dx} = C_1$$

$$\int dc = \int C_1 \, dx$$

$$\Downarrow$$

$$c = C_1 x + C_2$$

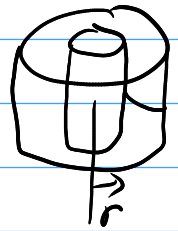
BC

$c(0,t) = c^0 \quad c(L,t) = c^L$

$$c(x) = c^0 - (c^0 - c^L)\frac{x}{L}$$

$$J = -D\frac{dc}{dx} = +D\frac{c^0 - c_L}{L}$$

② cylindrical



$$\nabla^2 c = \frac{1}{r}\frac{\partial}{\partial r}\cdot\left(r\frac{\partial c}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 c}{\partial \theta^2} + \frac{\partial^2 c}{\partial z^2}$$

B.C $c(r_1, \theta, z) = c_1$

$c(r_2, \theta, z) = c_2$ $\Rightarrow \frac{\partial^2 c}{\partial \theta^2} = \frac{\partial^2 c}{\partial z^2} = 0$

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{dc}{dr}\right) = 0$$

$$\int d\left(r\frac{dc}{dr}\right) = \int 0 \cdot dr$$

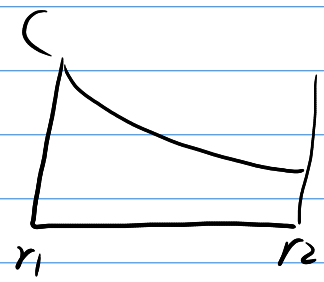
$$\Downarrow$$

$$r\frac{dc}{dr} = k_1$$

$$\int dc = \int \frac{k_1}{r}\, dr$$

$$\Downarrow$$

$$c = k_1 \ln r + k_2$$



Fit BC

$$k_1 \ln r_1 + k_2 = c_1$$

$$k_1 \ln r_2 + k_2 = c_2$$

$$k_1 \ln\frac{r_1}{r_2} = c_1 - c_2 \xrightarrow{\text{solve for } k_2}$$

$$k_1 = \frac{c_1 - c_2}{\ln\frac{r_1}{r_2}}$$

$$c(r) = c_1 - \frac{c_1 - c_2}{\ln\frac{r_2}{r_1}} \ln\left(\frac{r}{r_1}\right)$$

③ spherical $\quad C(r, \theta, \varphi)$

$$\nabla^2 C \Rightarrow \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial c}{\partial r}\right) + \underbrace{\boxed{}}_{\theta \text{ part}} + \underbrace{\boxed{}}_{\varphi \text{ part}} = 0$$

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2 \frac{dc}{dr}\right) = 0 \Rightarrow \int d\left(r^2 \frac{dc}{dr}\right) = \int 0 \, dr$$

$$r^2 \frac{dc}{dr} = k_1$$

$$\int dc = \int \frac{k_1}{r^2} dr$$

$$c = -\frac{k_1}{r} + k_2$$

B.C

$c(r = r_1) = c_1$

$c(r = r_2) = c_2$

$$\left.\begin{aligned} -\frac{k_1}{r_1} + k_2 &= c_1 \\ -\frac{k_1}{r_2} + k_2 &= c_2 \end{aligned}\right\} \left(\frac{1}{r_2} - \frac{1}{r_1}\right) k_1 = c_1 - c_2$$

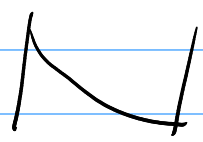
$$k_1 = \frac{c_1 - c_2}{\frac{1}{r_2} - \frac{1}{r_1}}$$

$$\left.\begin{aligned} -k_1 + r_1 k_2 &= c_1 r_1 \\ -k_1 + r_2 k_2 &= c_2 r_2 \end{aligned}\right\} \quad k_2 = \frac{c_1 r_1 - c_2 r_2}{r_1 - r_2}$$

$$c = \frac{c_1 - c_2}{\frac{1}{r_1} - \frac{1}{r_2}} \cdot \frac{1}{r} + \frac{c_1 r_1 - c_2 r_2}{r_1 - r_2}$$

$$= c_1 + (c_2 - c_1) \cdot \frac{\frac{1}{r} - \frac{1}{r_1}}{\frac{1}{r_2} - \frac{1}{r_1}}$$

Linear in $\frac{1}{r}$



④ Spatially varying D

We still have simple solution for $D \neq$ const

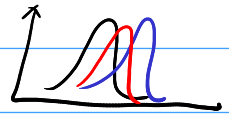
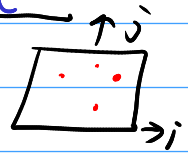
If D only depend on 1 variable (x) or (r)

$$\frac{d}{dx}\left(D(x)\frac{dc}{dx}\right) = 0 \xrightarrow{1^{st} \text{ int}} D(x)\frac{dc}{dx} = k_1$$

$$\int_{c_1}^{c} dc' = \int_{x_1}^{x} \frac{k_1}{D(x)} dx' \Rightarrow c(x) = c_1 + \underbrace{\int_{x_1}^{x} \frac{k_1}{D(x)} dx'}_{\text{Numerical int}}$$

Part II Non-steady state

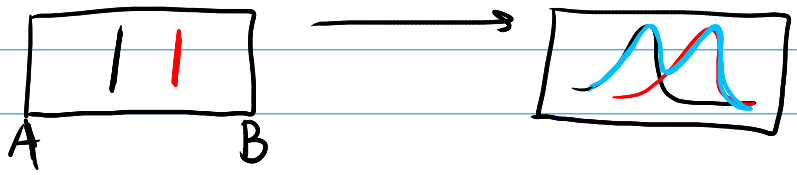
Different strategies exist

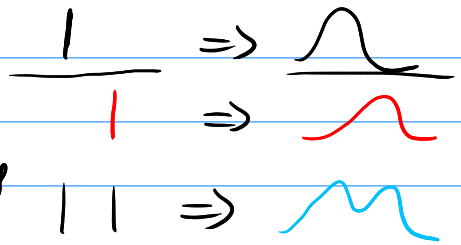
- Superposition + well-known source solutions 
- Laplace transform (real PDE → imaginary s-plane coeffs) 
- Numerical solution

Method 1

① Many cases use a well-know geometry

② If a geometry is complex ← addition of multiple terms (superposition)



Initial cond  ⇒ Solution a

⇒ Solution b

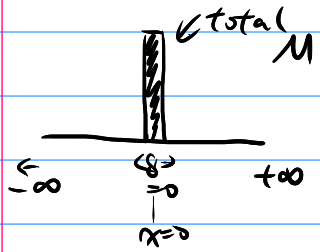
Combined source ⇒ Solution = a+b

Boundary conditions additive

In actual lecture talked later

Infinitely large system $\frac{dc}{dt} = D\frac{d^2c}{dx^2}$ ← Dirac delta func

$$C(x,0) = N\delta(x)$$



Solution

$$C(x,t) = \frac{K}{\sqrt{t}}\exp\left(-\frac{x^2}{4Dt}\right)$$

↑ Gaussian

Conservation $N = \int_{-\infty}^{+\infty} C(x)\,dx = \int_{-\infty}^{\infty}\frac{A}{\sqrt{t}}\exp\left(-\frac{x^2}{4Dt}\right)$

$$A = \frac{M}{2\sqrt{\pi D}} \qquad C(x,t) = \frac{N}{2\sqrt{\pi D t}}\exp\left(-\frac{x^2}{4Dt}\right)$$

How do we get there ? (Bonus part)

we have $\frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial x^2}$ $\begin{cases} \text{LHS - time} \\ \text{RHS - Coord} \end{cases}$

Recall $\sqrt{Dt}$ has length unit $\Rightarrow$ dimensionless transform

$$\eta = \frac{x}{\sqrt{Dt}} \Rightarrow \frac{\partial c}{\partial t} = \frac{\partial c}{\partial \eta}\cdot\frac{\partial \eta}{\partial t} = \frac{\partial c}{\partial \eta}\cdot\frac{x}{\sqrt{D}}\cdot\frac{1}{\sqrt{t^3}}\cdot -\frac{1}{2}$$

$t^{-\frac{1}{2}}$

$$= -\frac{1}{2}\frac{x}{\sqrt{Dt}}\cdot\frac{1}{t}\frac{\partial c}{\partial \eta}$$

$$= -\frac{1}{2}\eta\cdot\frac{1}{t}\cdot\frac{\partial c}{\partial \eta}$$

$$\frac{\partial^2 c}{\partial x^2} = \frac{\partial^2 c}{\partial(\eta\cdot\sqrt{Dt})^2} = \frac{1}{Dt}\frac{\partial^2 c}{\partial \eta^2}$$

$$\Rightarrow \frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial x^2} \Rightarrow -\frac{1}{2}\eta\frac{1}{\cancel{t}}\frac{\partial c}{\partial \eta} = \cancel{D}\cdot\frac{1}{\cancel{D}\cancel{t}}\frac{\partial^2 c}{\partial \eta^2}$$

$$\frac{\partial^2 c}{\partial \eta^2} + \frac{1}{2}\eta\frac{\partial c}{\partial \eta} = 0 \qquad \text{Let } u = \frac{\partial c}{\partial \eta}$$

$$\frac{\partial u}{\partial \eta} + \frac{1}{2}\eta u = 0 \qquad \frac{\partial u}{u} = -\frac{1}{2}\eta\,\partial\eta$$

$$\int\frac{du}{u} = \int -\frac{1}{2}\eta\,d\eta \Rightarrow \ln u = -\frac{1}{4}\eta^2 + \text{Const}$$

$$u = C_1\exp(-\tfrac{1}{4}\eta^2)$$

since $u = \frac{\partial c}{\partial \eta}$

$$\frac{\partial c}{\partial \eta} = C_1\exp(-\tfrac{1}{4}\eta^2) \Rightarrow \int dc = \int C_1\exp(-\tfrac{1}{4}\eta^2)\,d\eta$$

$$c(\eta) = \overset{\text{coeff}}{\square} + \overset{\text{coeff}}{\square}\underbrace{\int\exp(-\tfrac{1}{4}\eta^2)\,d\eta}$$

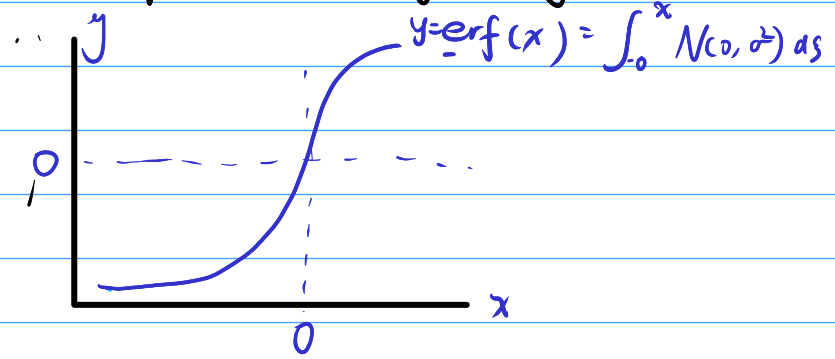
error function form

$$\operatorname{erf}(\xi) = \frac{2}{\sqrt{\pi}}\int_0^{\xi} e^{-x^2}\,dx$$

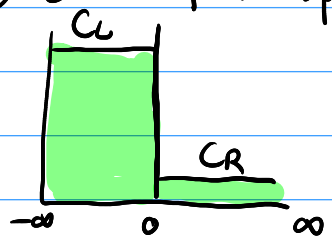
$$= k_1 + k_2\operatorname{erf}\left(\frac{\eta}{2}\right) = \boxed{k_1 + k_2\operatorname{erf}\left(\frac{x}{\sqrt{4Dt}}\right)}$$

k_1 & k_2 are determined by the initial & B.C

Error function: integral of gaussian

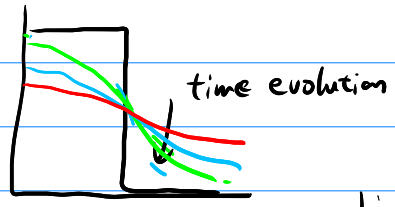


① Semi-infinite space



$$C(\eta) = \frac{C_L + C_R}{2} + \frac{C_R - C_L}{2}\,\mathrm{erf}\left(\frac{\eta}{2}\right)$$

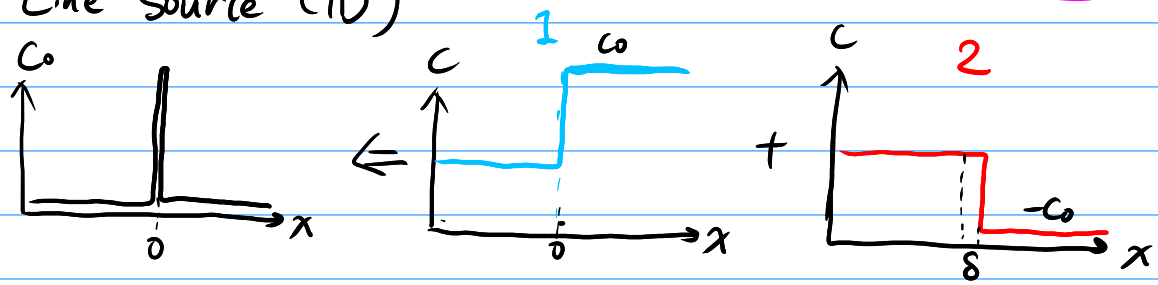
$$C(x,t) = \bar{c} + \frac{\Delta C}{2}\,\mathrm{erf}\left(\frac{x}{\sqrt{4Dt}}\right)$$



what if $t \to \infty$?

$$\lim_{t\to\infty} \mathrm{erf}\left(\frac{x}{\sqrt{4Dt}}\right) = 0 \qquad C(x,\infty) = \bar{c}$$

② Line Source (1D)



Line source is combined 2 semi-infinite sources!

Superposition $\longrightarrow$ $C(x,t) = \underbrace{C_1(x,t)}_{\text{Solution for 1}} + \underbrace{C_2(x,t)}_{\text{Solution for 2}}$

$$\left.\begin{aligned} C_1 &= \frac{C_0}{2} + \frac{C_0}{2}\,\mathrm{erf}\left(\frac{x}{\sqrt{4Dt}}\right) \\ C_2 &= -\frac{C_0}{2} + -\frac{C_0}{2}\,\mathrm{erf}\left(\frac{x-\delta}{\sqrt{4Dt}}\right) \end{aligned}\right\} \Rightarrow C(x,t) = \frac{C_0}{2}\left[\mathrm{erf}\left(\frac{x}{\sqrt{4Dt}}\right) - \mathrm{erf}\left(\frac{x-\delta}{\sqrt{4Dt}}\right)\right]$$